

## BINLEVEL · REQUIREMENTS

# BinLevel requirements

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## DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
| Document ID | BNL-REQ-001                           | Version | 0.3            |
| Status      | <b>Draft</b>                          | Date    | 2026-09-25     |
| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept         |         |                |
| Repository  | github.com/BoujeeEnjinia1701/binlevel |         |                |
| Source      | docs/03-requirements.md @ ab14f8a     |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                 |
|---------|------------|--------------|------------------------------------------------------------------------------------------------------------------------|
| 0.3     | 2026-09-25 | Amish Chadha | Status checked by calculation (BNL-CAL-001); adopted defaults from BNL-DDR-001 written into R3 and R9; 11-byte payload |
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept                                               |
| 0.1     | 2026-09-25 | Amish Chadha | Initial scaffold                                                                                                       |

## BinLevel requirements

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These requirements are the TRL 3 design basis. The design choices behind them are adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review (BNL-DDR-001); no target was relaxed or redefined. Status is from the sizing note BNL-CAL-001, which checks each requirement against the parametric model: eight are met (four by calculation, four by design), three are at risk (R2, R6, R9), three are not met (R5, R10 in steel containers, R16 on mass) and two can only be settled by test (R7, R12). See the notes under Table 1.

Table 1. Requirements.

| ID  | Requirement                                                                            | Target                                                                                                                                             | Verification (TRL 3 or later)                                     | Status at TRL 3 (BNL-CAL-001)                                                                                                                                                                    |
|-----|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Measure distance from lid to waste surface over the full depth of the target container | 0.03 to 1.5 m below the sensor face                                                                                                                | Calculation and datasheet review; later bench test in a container | Met: 0 to 1.01 m needed; ultrasonic from 0.25 m, ToF to 0.4 m, overlapping between 60 % and 75 % fill (datasheet ranges to confirm)                                                              |
| R2  | Report fill level accurately on real mixed waste                                       | Within $\pm 10$ % of container depth for 90 % of readings                                                                                          | Error budget; later field comparison against manual dip readings  | <b>At risk:</b> instrument error about 10 mm (ultrasonic) and 20 mm (ToF) of the $\pm 101$ mm allowance; the waste surface is unknown                                                            |
| R3  | Report often enough to plan routes, and quickly when full                              | Routine uplink at least every 2 h (default 1 h, 2 h at SF11 and SF12); threshold alert within 20 min of crossing the set fill level (default 80 %) | Timing calculation                                                | Met: worst-case alert 17.5 min                                                                                                                                                                   |
| R4  | Run for years on one battery                                                           | 5 years or more in the worst radio case (SF12); 10 years design target                                                                             | Power budget calculation                                          | Met: 16.3 years on a C cell at SF12; enclosure and seals, not the cell, limit life                                                                                                               |
| R5  | Survive rain, condensation and washing                                                 | IP67 minimum; IP69K target where containers are pressure washed                                                                                    | Enclosure datasheet; later spray test                             | <b>Not met for IP69K:</b> stock IP67 box (DDR-001, D9)                                                                                                                                           |
| R6  | Operate across the climate range                                                       | -20 to +60 °C inside the container                                                                                                                 | Heat balance and datasheet review                                 | <b>At risk:</b> a dark lid reaches about 67 °C on a 40 °C day; the standard module is rated only down to -20 °C; transducer rating and condensation unverified                                   |
| R7  | Survive lifting, tipping and lid slams                                                 | Repeated shocks of the emptying cycle without loosening; target IK08 impact on the enclosure                                                       | Load calculation; later drop and shake test                       | Not verifiable at TRL 3: 85 N at 20 g against about 4,200 N per bolt; loosening and impact need a test                                                                                           |
| R8  | Detect and report emptying and lid events                                              | Tip event from the accelerometer reported with the next uplink                                                                                     | Firmware logic review                                             | Met by design                                                                                                                                                                                    |
| R9  | Warn of abnormal heat in the bin                                                       | Alert uplink within 15 min when temperature exceeds 70 °C or rises more than 15 K in 15 min                                                        | Timing calculation and heat balance                               | <b>At risk:</b> alert within 7.5 min with temperature read every 5 min, but sun breaking through cloud raises the unit 15.4 K in 15 min (false alarm); maintenance aid only, not a fire detector |
| R10 | Connect over open LoRaWAN within radio rules                                           | LoRaWAN 1.0.x Class A; EU868, US915 and IN865 plans; within duty cycle and TTN's 30 s per day fair use                                             | Airtime and link calculation; later range test                    | <b>Not met in steel containers</b> with the internal antenna (about 0.35 km at SF12); met in HDPE containers: 20.8 s per day at SF12, 11-byte payload fits US915 DR0, about 0.8 to 1.3 km range  |

| ID  | Requirement                                    | Target                                                                                            | Verification (TRL 3 or later)                  | Status at TRL 3 (BNL-CAL-001)                                                                     |
|-----|------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------|
| R11 | Protect privacy                                | No camera or microphone; payload limited to fill, distance, temperature, tilt, events and battery | Design review of BOM and payload               | Met by design                                                                                     |
| R12 | Install without wiring or tools beyond a drill | Fit to a lid in 10 min or less with four bolts or rivets; no cables                               | Timed trial at TRL 4                           | Not verifiable at TRL 3: four through-bolts on a 130 x 60 mm pattern, no cables                   |
| R13 | Resist casual tampering and theft              | Mounted inside the lid; tamper-resistant fasteners                                                | Design review                                  | Met by design                                                                                     |
| R14 | Low cost and buildable                         | Parts cost \$60 or less per sensor at prototype quantities; no custom tooling                     | Priced BOM                                     | Met: \$55.00 (indicative); \$63.00 with the external antenna option                               |
| R15 | Open and interoperable                         | Documented payload format and open decoder; works with any LoRaWAN network server                 | Documentation review                           | Met by design: candidate 11-byte layout in BNL-CAL-001                                            |
| R16 | Light and compact                              | 400 g or less; no larger than 160 x 90 x 100 mm below the lid, including bracket                  | Mass calculation from the model, then weighing | <b>Not met on mass:</b> about 435 g with a 1.5 mm stainless bracket; size met at 150 x 80 x 61 mm |

Notes on requirements not met or at risk:

- **R2 at risk.** Ultrasonic echoes from bags, boxes and bulky items are uneven, and the waste surface is rarely flat. The mitigation is to take the median of several pings and to combine the two sensors, but accuracy on real waste is unknown until measured.
- **R5 not met for IP69K.** A low-cost stock enclosure is typically rated IP67. Where containers are washed with hot pressure water, a sealed potted design or an IP69K enclosure is needed, which raises cost.
- **R6 at risk.** The requirement's 60 °C upper limit is below the about 67 °C a dark lid reaches in hot sun. Raising the limit to 70 °C and naming the RAK3172-T (-40 to 85 °C) are proposed, awaiting Amish; the target is not changed here.
- **R9 at risk.** The 15 K rate-of-rise trigger can fire when sun breaks through cloud. A revised rule (rate of rise counted only above 50 °C, or the absolute threshold raised) is proposed, awaiting Amish; the target is not changed here.
- **R10 not met in steel containers.** An antenna inside a steel container loses about 30 dB (assumed). HDPE containers are fine within about 1 km of a gateway. The external antenna or the exclusion of steel containers is open (DDR-001, O1).
- **R16 not met on mass.** The stainless bracket alone is 183 g. A 2 mm aluminium bracket (unit about 334 g) or a relaxed limit is proposed, awaiting Amish.

## ■ Assumptions

- Target container: 1,100 L four-wheel communal container (DDR-001, D1), lid underside 1.07 m above the inner floor and transducer face 1.01 m above it (from the parametric model).

- Fill level is reported as a percentage of the distance from the sensor face to the container floor, measured at installation.
- Worst radio case is spreading factor 12 at 125 kHz with an 11-byte payload (1.48 s airtime per uplink); typical urban case is SF9 (0.21 s).
- A dark lid may reach about 67 °C on a 40 °C day in full sun (BNL-CAL-001, section H; to be measured).